

MARANDA 2025 KCSE PREDICTION



EXPECTED EXAM 1 COMPUTER STUDIES

451/1

PAPER 1 (Theory)

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- Write your Name and Index Number and Adm No in the spaces provided.
- This paper consists of TWO sections: A and B
- Answer **ALL** questions in section A
- Answer question **16** and any other **THREE** questions from section **B**
- **All** answers should be written in the spaces provided in the question paper

For Examiners use only.

Section	Question	Candidates Score
A	1 – 15	
B	16	
	17	
	18	
	19	
	20	
Total Score		

SECTION A (40 marks)

Answer all the questions in this section.

1. Explain what is meant by the terms:- (4 marks)

i) FDD.....
.....

ii) HDD.....
.....

iii) CMOS.....
.....

iv) TFT.....
.....

2. Outline TWO application areas of Artificial Intelligence (2 marks)
.....
.....
.....

3. State TWO ways of identifying an illegal copy of an Operating System (2 marks)
.....
.....
.....

4. List any four publications that can be created using a Publishing application software (2 marks)
.....
.....
.....

.....

5. Describe a speech synthesis system

(2 marks)

.....

6. Find the Value of T_8 in the following binary equation:

$110_{10} - AB_{16} = T_8$

(3 marks)

.....

7. State any TWO factors one would consider when selecting any data output device. (2 marks)

.....

8. Input devices can be categorized according to how they are used to enter data namely keying devices, pointing devices, scanning devices, speech recognition devices and other digital devices. Give TWO examples each of the following devices.

(i) Scanning devices.

(1 mark)

a).....(b).....

(ii) Other digital devices.

(1 mark)

a).....(b).....

(iii) Pointing devices.

(1 mark)

a).....(b).....

9. Describe the following types of monitors

(3 marks)

a) Cathode Ray Tube(CRT)

.....

.....

b) Liquid Crystal Display (LCD)

.....

.....

c) Gas Plasma

.....

.....

10(a) State TWO functions of Read Only Memory (ROM)

(2 marks)

.....

.....

.....

(b) Give two characteristics of ROM

(2 marks)

.....

.....

.....

11. Explain the meaning of the following system characteristics.

(2 marks)

i) Entropy

.....

.....

ii) Controls

.....

.....

12. Discuss TWO computational errors giving example in each case. (4 marks)

.....
.....
.....
.....
.....
.....

13. Differentiate between WORM and a VIRUS. (2 marks)

.....
.....
.....
.....

14. Differentiate sequential file organization from indexed sequential file organization. (2 marks)

.....
.....
.....
.....

15. Give any THREE functions of the central processing unit (CPU) (3 marks)

.....
.....
.....
.....

SECTION B (60 marks)

Answer questions 16(Compulsory) and any other three questions from this section.

16. a) Differentiate between Assembler and Interpreter as used in programming. (2 marks)

.....

.....

.....

.....

.....

b) Give any THREE objectives that fourth génération programming languages were designed to meet. (3mks)

.....

.....

.....

.....

.....

c) On the Nairobi-Nakuru highway, the Kenya Police have put speed cameras at a certain point to read the time a vehicle passes a point (A) on the road and then reads the time it passes a second point (B) on the same road. (Points A and B are 200 meters apart). The speed of the vehicle is calculated using:

- $Speed = \frac{100}{(time\ at\ point\ B - time\ at\ point\ A)} (Km/hr)$
- *The maximum allowed speed is 100 kilometers per hour.*
- *500 vehicles were monitored using these cameras over a 1-hour period.*

(i) Write a pseudo code, which: (5 marks)

- Inputs the start time and the end time for the 500 vehicles that were monitored
- Calculates the speed for each vehicle using the formula above.
- Outputs the speed for each vehicle and a message if the speed exceeded 100 km/hour.

≥ 100 km/hr “High Speed”

<100km/hr “Normal Speed”

PSEUDOCODE

[illegible]

[illegible][illegible]

17.(a) Distinguish between the following sets of terms as used in spreadsheets

(i) Worksheet and workbook. (2 marks)

(ii) Filtering and sorting. (2 marks)

(b) State one way in which a user may reverse the last action taken in a spreadsheet package. (1mark)

(c)The following is a sample of a payroll. The worksheet row and column headings are marked 1,2,3..... and A,B,C..... respectively.

	A	B	C	D	E	F	G	H
1	NAME	HOURS WORKED	PAY PER HOUR	BASIC PAY	ALLOWANCE	GROSS PAY	TAX DEDUCTION	NET PAY
2	WAITHERA	21	750					
3	CHEPKOECH	28	850					
4	CHUMBA	14	550					
5	NKIROTE	20	1000					
6	NANA	23.5	250					
7	KIBET	25.5	500					
8	AUMA	14	250					
9	ATITWA	18	1500					
10	MOCHAMA	35	350					
	TOTALS							

Use the following expressions to answer the questions that follow

- Basic pay = Hours worked x pay per hour
- Allowances are allocated at 10% of basic pay
- Gross pay = Basic pay + allowances
- Tax deductions are calculated at 20% of gross pay.
- Net pay = Gross pay – tax deductions.

Write formulae using cell references for the following cells:

(i) D2..... (1mark)

- (ii) E4.....(1mark)
- (iii) F10.....(1 mark)
- (iv) G7.....(1 mark)
- (v) H5..... (1 mark)

- d) (i) State THREE ways of moving round a page in Desktop Publishing window. (3 marks)
- ii) State TWO ways on how Information and Communication Technology (ICT) has been used to reduce the spread of COVID 19 (2 marks)

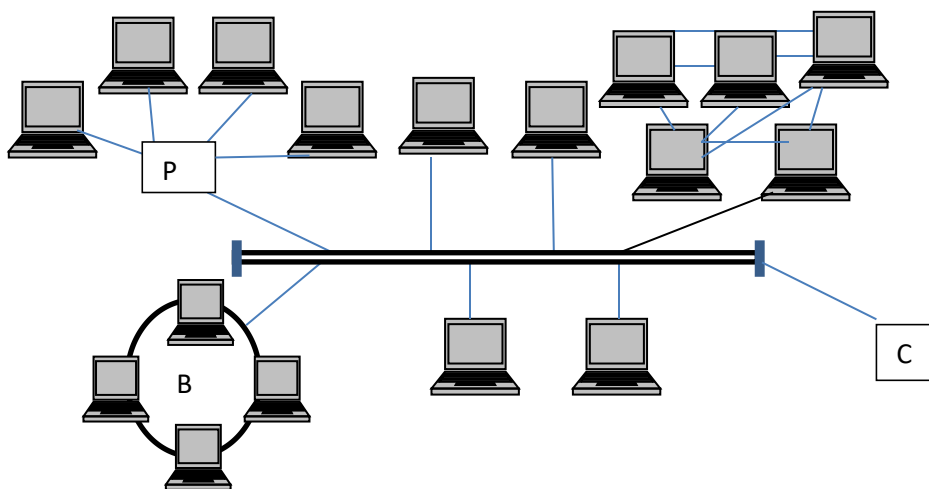
18. a.(i) List down THREE advantages of ATM cards. (3 marks)

- (ii) State FOUR components of an electronics Point of Sale. (2 marks)

- b. Give THREE reasons why a manager may prefer to teleconference during meetings. (3 marks)
- c. Menengai Company would like to introduce use of a new information system. State THREE reasons why the company would like to introduce a new information system? (3 marks)
- d. Once the whole system has been designed and constructed, it will need to be implemented. Describe TWO system changeover strategies that are likely to be used. (4 marks)

- 19.a) Explain why an intranet is more secure when sharing files within an organization compared to the internet. (1 mark)

- b) The network setup below is formed by combining several LANS. Use it to answer the questions (i) – (v) below.



- (i) Identify the network device P and C. (1mark)
- (ii) State the importance of the network device P and C (2 marks)
- (iii) What is the name given to the combined topology in figure above? (1mark)

(iv) Identify the network topologies used in the above diagram **(2marks)**

(v)State TWO disadvantages of topology connected to device P. **(2marks)**

c) A company has three branches M,N and O , where M is the headquarter. The local area network (LAN) at M is directly connected to LAN at N and directly connected to the LAN at O. **Explain two** benefits of having LAN at N also directly connected to the LAN at O. **(4 marks)**

d) The government of Kenya ordered all broadcasting TV stations to switch from analog transmission to digital TV transmission. **State two** benefits of digital transmission over analog transmission **(2 marks)**

20. a) Describe the following computer memories. **(2 marks)**

i. Register:

ii. Cache:

b. Explain Real-time Processing data processing mode giving TWO areas where it can be used. **(5 marks)**

c. Describe any TWO coding schemes used in Data representation **(4 marks)**

d. List FOUR devices controlled by the Operating System **(2 marks)**

e. Define a Laptop **(2 marks)**